

SQL Fundamentals:

SELECT and Filtering

① Selecting and Sorting

② SELECT *

FROM employees-table;

employees

id	first_name	last_name	Department	Salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	Daniel	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

③ Find all unique departments

SELECT DISTINCT department
FROM ~~dept~~ employees-table;

department
IT
HR
Finance
Marketing

④ Retrieve first and last ordered by salary descending

SELECT first_name,
last_name

FROM employees table
ORDER BY salary DESC;
employees

first-name	last-name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

④ Retrieve the top 3 highest-paid employees.

SELECT first-name,
last-name,
salary,
id

FROM ~~employees~~ employee
ORDER BY salary DESC
LIMIT 3;

employees

id	first-name	last-name	salary
5	Eva	Naidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

② Filtering with WHERE

⑤ Find employees in the IT department

SELECT id, first-name, last-name, department
FROM employees
WHERE department = 'IT';

employees			
id	first-name	last-name	department
1	Alice	Green	IT
5	Eva	Niadoo	IT

⑥ Find employees in Finance with Salary > 60000

```
SELECT id, first-name, last-name, department
FROM employees
WHERE salary > 60000
AND department = 'Finance';
```

id	first-name	last-name	department	Salary
3	Cathy	Zulu	Finance	65000

⑦ Find employees in HR or Marketing

```
SELECT id, first-name, last-name, department
FROM employees
WHERE department = 'HR' OR 'Marketing';
```

id	first-name	last-name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

⑧ Find employees not in IT.

```
SELECT id, first-name, last-name, department
FROM employees
WHERE department NOT 'IT';
```

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

Q9 Find employees in IT, HR, or Finance using IN

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');
```

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance

Q10 Find employees in IT with salary > 65000 and City Johannesburg

```
SELECT id, first_name, last_name, department, salary, city
FROM employees
WHERE department = 'IT'
AND salary > 65000
AND city = 'Johannesburg';
```

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70 000	Johannesburg
5	Eva	Naidoo	IT	72 000	Johannesburg